Spatial Cluster Randomized Trials - Sampling Design with Spillover Effects & Spatial Dependence

Andrew Walther[1], Tonya Van Deinse[2], and Feng-Chang Lin[1]

[1]The University of North Carolina at Chapel Hill, Department of Biostatistics

[2]The University of North Carolina at Chapel Hill, School of Social Work

*Corresponding author: flin@bios.unc.edu

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Abstract

Background: Investigators designing cluster randomized trials desire insights directing treatment assignment methodology for studies involving spillover effects and spatial dependence.

Methods & Simulation: Treatment assignment strategies including simple random sampling (SRS) and block stratified sampling (BSS) are defined and spatial autoregressive modeling is applied with consideration for spillover effects and spatial dependence for estimation of intervention effects. A simulation study is carried out comparing SRS and BSS sampling methods on spatial grids of varying sizes. A range of spillover effects and levels of spatial dependence were considered for estimation of the intervention effect via a spatial autoregressive (SAR) model.

Results: Findings of an extensive simulation study comparing simple random sampling and block stratification methods indicate that randomly selected treatment assignments result in best case reduced Mean Squared Error (MSE) when estimating intervention effects, but block stratified treatment assignments ~~lead to minimal~~ generally lead to less variation in MSE among a series of treatment combinations, indicating that a block stratified treatment arrangement will not achieve the minimal level of estimation error, but it remains robust across a range of selected parameters. While

SRS achieves lower average MSE in certain scenarios, both designs exhibit substantial estimation bias when treatment and spillover indicators are collinear. Variation in performance across all possible treatment combinations and the worst-case allocation risk this entails motivates consideration of BSS as a more consistent alternative design.

Conclusion: The relationship between spillover effects and mean squared errors (MSE) of intervention effect estimation is apparent. The MSE for the intervention effect, which is the average MSE over each of N simulation iterations, is minimized for some combinations of random sampling treatment assignment, but block stratified assignment ~~minimizes~~ generally reduces variance among combinations of possible treatment arrangements. In short, SRS may achieve lower average MSE in some scenarios, but BSS bounds worst-case allocation risk through ~~near-zero~~ lower variation in estimation error across treatment combinations; the preferred strategy depends on the anticipated spillover mechanism, with BSS most advantageous when spillover is expected to flow primarily into control clusters.

1 Introduction

1.1 Background & Motivation

Randomized controlled trials (RCTs) represent the gold standard for assessing the efficacy and safety of new interventions[1][2]. However, in many real-world scenarios, particularly in public health, education, and the social sciences, randomizing individuals is not feasible due to logistical constraints, ethical considerations, or the inherent nature of the intervention itself, which may be administered at a community level. In such cases, the cluster randomized trial (CRT) emerges as the preferred and most rigorous design, where intact social units or groups such as villages, schools, or clinics are randomized to intervention or control arms[3][4].

A key advantage of the CRT design is its ability to evaluate interventions where there is high risk of contamination between treated and untreated individuals within the same cluster[3][5]. Furthermore, CRTs allow estimation of not only the direct effects of an intervention on those who receive it but also the indirect spillover effects on untreated individuals, typically those in control clusters. Measuring these spillover effects is often of primary scientific and policy interest, as it provides a comprehensive assessment of an intervention's total public health impact[3][6]. When trial clusters are defined by geographic areas,

as is frequently the case, the design must contend with inherent spatial complexities that can challenge the core statistical assumption of cluster independence. Two primary and interrelated spatial phenomena can profoundly influence trial outcomes and compromise the validity of the results: spillover effects and spatial dependence[3][2].

Spillover occurs when an intervention's effects in one cluster extend or "spill over", into neighboring clusters, creating significant analytical challenges[7][8]. This is especially common in trials of infectious disease interventions. A study on the use of permethrin-impregnated bed nets by Binka et al. (1998) provided early, compelling evidence of this phenomenon, demonstrating that child mortality was significantly lower in control villages situated near intervention villages, presumably due to a reduction in the local mosquito population[9][10][11]. Similarly, the work by Miguel and Kremer (2004) on a school-based de-worming program in Kenya found significant positive externalities, showing that the health and school attendance of untreated children improved if they lived near schools that were part of the intervention[12]. The critical implication is that failing to account for such positive spillover can lead to a substantial underestimation of the intervention's true effect, as the measured outcomes in the control group are artificially improved by their proximity to the intervention arm. Compounding this issue is spatial dependence, a fundamental concept supported by Tobler's First Law of Geography that asserts nearby observations tend to be more related than distant ones[8][13]. In the context of a CRT, this principle suggests that outcomes in nearby clusters may be correlated due to unmeasured, spatially-patterned environmental, social, or demographic factors. The presence of this spatial autocorrelation violates the assumption that clusters are independent units, which can lead to biased standard errors and an increased Type I error rate, undermining statistical inference[14][15][13].

In response to these challenges, a body of literature has emerged focusing on post-hoc analytical adjustments to account for spatial effects after the data have been collected. Jarvis et al. (2017) classifies these approaches into two categories: the spatial variable approach, which constructs a new covariate from distance or density metrics, and spatial modeling[8]. Examples include measuring the straight-line distance from control participants to the nearest intervention household (Hawley et al., 2003 & Gimnig et al., 2003)[11][10], calculating outcome rates across different distance bands (Binka et al., 1998)[9], or quantifying the density of treated individuals within a predefined radius (Miguel & Kremer, 2004 & Lenhart et al., 2008)[12][16]. More recently, Chao et al. (2015) developed a "potential

exposure" metric to quantify the spatially heterogeneous risk surrounding an individual[17]. In contrast, the spatial modeling approach is more statistically sophisticated, directly incorporating the geographic structure of the data into the random effects component of a statistical method. The method employed by Alexander et al. (2003)[18] considers a distance-decay function in the covariance matrix and an alternative by Silcocks and Kendrick (2010)[19] using spatial weights matrices treats spatial correlation as an underlying unobserved process. Watson & Smith (2025) propose a distinct framework in which treatment is measured as a continuous dose based on distance to the nearest intervention site, estimating a dose-response function (DRF) rather than a discrete cluster-level intervention or spillover effect[20].

Despite the growing sophistication of these analytical tools, they are fundamentally post-hoc corrective solutions to a problem that could (and should) be addressed proactively at the design stage. The existing literature reveals a noticeable lack of research into how the initial sampling and allocation of treatments across a geographic landscape can be optimized to manage these known spatial complexities. The review by Jarvis et al. (2017) highlights this gap, noting that in practice, the impact of location on trial results is definitely an afterthought and oftentimes entirely ignored[8]. In fact, trial designs explicitly intended to mitigate spatial effects, such as creating well-separated clusters or buffer zones, were so uncommon that studies employing them were excluded from their primary analysis, underscoring how peripheral these considerations have been to standard practice[8].

This oversight has significant consequences because known studies that compare spatial analytical approaches consistently find that accounting for spatial effects is important. Silcocks and Kendrick (2010) demonstrated that spatial models not only fit primary care trial data better than standard CRT models, but also that their inclusion altered both the intervention effect point estimate and its precision[19]. Similarly, Chao et al. (2015) found that adjusting for their spatial exposure metric lowered the estimated effect of a typhoid vaccine[17]. The logical conclusion is that standard trial designs that rely on the simple random allocation of clusters may be systematically producing biased or inefficient estimates without investigators being aware. This motivates the central question of this study: "How does the choice of an *a priori* treatment assignment sampling strategy impact the estimation of intervention effects in the presence of both spatial spillover effects and spatial dependence?" While we have methods to analyze the problem after a study is completed, there is currently a lack of evidence-based guidance for investigators on how to design a cluster randomized trial to mitigate these issues from the outset of study planning.

1.2 Aims & Contributions

This study aims to address this critical gap by moving beyond post-hoc analysis to investigate the important role of treatment sampling design in spatial CRTs. Through a comprehensive simulation study, we systematically compare the performance of two contrasting approaches for assigning study clusters to intervention and control arms: simple random sampling (SRS), representing the standard practice for treatment assignment, and a block stratified sampling (BSS) method. The BSS strategy is intentionally designed to create spatial separation between clusters of the same treatment assignment by isolating them to potentially reduce the confounding influence of spillover and spatial dependence.

Our simulations explore how the estimation error of the intervention effect, quantified by Mean Squared Error which is composed of bias and variance, varies across a range of realistic conditions including different magnitudes of spillover effects, the presence or absence of underlying spatial dependence, and variation of restrictions on the geographic distance over which spillover can occur. By identifying the systematic patterns that emerge, this novel research provides practical, evidence-based guidance for investigators designing future spatial CRTs. The findings will equip researchers with the knowledge to select sampling strategies that align with their specific study goals, ultimately leading to more robust, efficient, and valid estimates of intervention effects in complex spatial settings.

The remainder of this article is organized as follows: Section 2 outlines the theoretical framework of the study, including the data generation process, treatment sampling procedures, and estimation methods employed. Section 3 details the procedure and results of the simulation study undertaken to evaluate the sampling methods considered. Section 4 considers an application of these findings to make a recommendation for other investigators. Finally, Section 5 discusses the implications of our findings, potential drawbacks to the work performed, and potential areas for continued future research.

2 Design of Spatial CRTs

2.1 Cluster Randomized Trials

The randomized controlled trial (RCT) serves as the gold standard design to evaluate intervention efficacy[21]. Its core principle, the random allocation of individual participants, is expected to balance both measured and unmeasured confounding variables across treatment arms[22]. However, individual

randomization is often impractical, or an intervention is delivered at a group level, necessitating an alternative approach[4][3]. In these scenarios, the cluster randomized trial (CRT) is employed, where intact social units or groups like households, clinics, or geographic areas are randomized to intervention arms[4][23]. A key advantage of this design is the limitation of treatment contamination, where control group members are inadvertently exposed to the intervention, which would otherwise bias the estimated treatment effect toward the null[3]. Statistical considerations in a CRT are the measured response from an intervention and the inherent correlation among outcomes of individuals within the same cluster, a dependency quantified by the intraclass correlation coefficient (ICC)[4][24]. The ICC measures how similar the outcomes of individuals within a cluster are likely to be, relative to those of other clusters. This intra-cluster correlation violates the standard assumption of independence between observations. The failure to account for this feature in the analysis leads to underestimated standard errors and an inflated probability of a Type I error[25][26]. Therefore, the valid design and analysis of CRTs require specialized statistical methods that properly account for this clustering structure, with comprehensive frameworks provided by Donner & Klar (2000)[4] and Hayes & Moulton (2017)[3].

2.2 Neighbor Definitions

2.2.1 Contiguity-Based

Contiguity-based neighbors[27] are constructed with the assumption that neighbors of a given area (individual cells in the case of this study) are other clusters (or cells) that share a common boundary. This boundary may be an edge and/or vertex. The **spdep** R package[28][29][30][31] includes functions for working with spatial neighbors and spatial dependence structures. In this study, we consider Rook neighbors (shown in Figure 1 (a)), which fulfill a contiguity condition where a pair of clusters share a common boundary made up by an edge. Queen neighbors (shown in Figure 1 (b)) are another type of contiguity-based neighbors that fulfill a contiguity condition where a pair of clusters share a common boundary of either an edge and/or a vertex[27]. Clusters may also have Bishop neighbors where a common boundary at the vertex (corner) of the spatial region is shared. Clusters may also have higher-order neighbors where they do not directly share an edge or vertex, but rather have a common boundary with a cluster that is a neighbor of another cluster. In other words, these are “neighbors of neighbors” of the reference cluster.

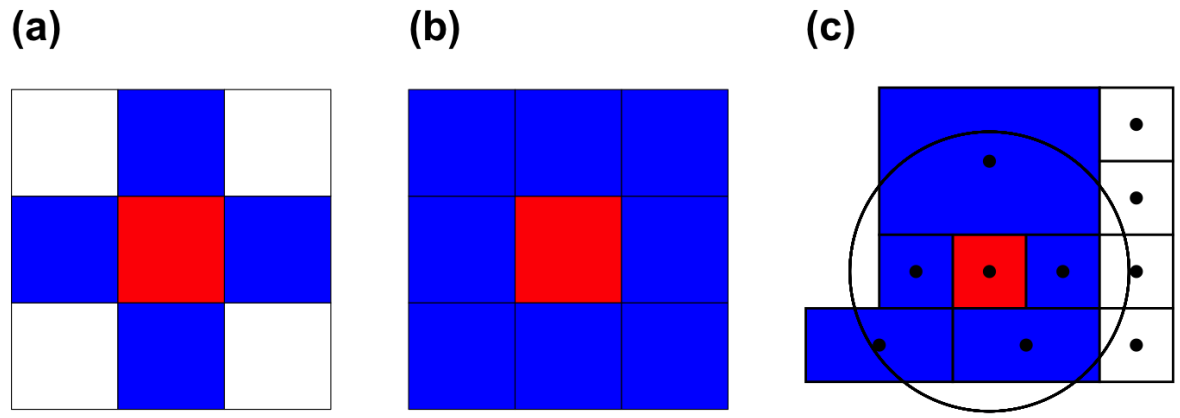


Figure 1: (a) Rook neighbors (blue) share an edge with a reference spatial region (red), (b) Queen neighbors (blue) share either an edge or vertex (or both) with a reference spatial region (red), and (c) Distance-based neighbors (blue) have centroids within a fixed radius of the reference cluster (red). Clusters with centroids beyond this distance are not considered as neighbors (white)[32]

2.2.2 Distance-Based

Distance-based neighbors[27] identify as neighbors any cluster whose centroid falls within a specified radius of a reference cluster (illustrated in Figure 1 (c)). A complete description of distance-based neighbor construction and the corresponding spatial weights formulation is provided in Appendix D. The simulation study presented in this paper relies exclusively on Rook contiguity for neighbor definition.

2.3 Intervention & Spillover Effects

Two primary effect types are considered to understand how treatments affect subjects in clusters where they are applied and subjects in the surrounding areas. The intervention effect and spillover effect are two parameters that are estimated in this study. Spatial cluster randomized trials have subjects or regions that are assigned to receive an intervention and others assigned as controls that do not receive an intervention. Observation of the resulting outcomes allows for estimation of the direct intervention effect and indirect spillover effect on the study region.

2.3.1 Intervention Effect

The intervention effect (β) is the *direct* structural coefficient of the intervention in the spatial autoregressive model of Section 2.5: the change in a cluster's outcome attributable to its own receipt of the

intervention, holding the spillover indicator z_i and the spatial lag fixed. Estimation of this coefficient is possible due to the presence of clusters assigned as controls for comparison. The relevant intervention could be a vaccine to prevent infectious disease, a training program for government employees, an educational program for students, or many others. When spillover propagates back into intervention clusters (the Control & Intervention mechanism; see Section 2.3.3), β remains the direct coefficient, and the spillover received by treated units is carried separately by the coefficient ψ on z_i rather than being absorbed into β.

2.3.2 Spillover Effect Methods

The spillover effect (ψ) is the structural coefficient on the spillover indicator z_i in the SAR model (Section 2.5), quantifying the model-estimated change in outcome for a cluster that is adjacent to at least one intervention cluster, an indirect (spillover) effect on clusters where the intervention was not directly applied[33][34]. In general, a spillover effect can be specified as a function of spatial proximity to the locations where an intervention is applied, for example as a decay function of the distance from a unit to the nearest intervention cluster, or as a function of the number of neighboring intervention clusters to which a unit is exposed. In this study, however, spillover is modeled as a single binary cluster-level exposure: the indicator $z_i \in \{0, 1\}$ equals 1 when cluster i shares a Rook edge with at least one intervention cluster (and the spillover type permits it) and 0 otherwise. Because z_i is a binary 0/1 indicator rather than a distance- or count-valued quantity, the associated parameter ψ is a single scalar multiplier rather than a function. Because z_i is defined at the cluster level, all subjects within a cluster share the same spillover exposure, so spillover does not vary across individuals within a cluster. Richer specifications that allow spillover to decay with distance or to accumulate over multiple treated neighbors are natural extensions, discussed in Section 5, but are beyond the scope of the present study.

A consequence of representing spillover through a binary cluster-level indicator is that, on the small grids considered here, z_i can be strongly correlated with the treatment indicator x_i, and for some allocations the two are exactly collinear, in which case β and ψ are not separately identifiable from the model fit alone. The conditions under which this occurs and its consequences for estimating β are examined in Section 3.

2.3.3 Spillover Effect Types

In this study, there are two specific ways in which spillover of the applied intervention is allowed to propagate to other cells or subjects. In both cases, spillover effects originate from areas where the original intervention treatment is applied. The first case allows for spillover to propagate from cells receiving the intervention into only cells assigned as control cells. In this case, the effect of the intervention is not permitted to spillover into other intervention cells. This suggests a case where the intervention effect applied is the maximum observable effect and any potential spillover cannot increase the observed effect of a particular intervention. Another method of spillover application is considered where spillover effects are permitted to be applied to both control AND intervention cells that are neighbors of a cell that receives an intervention. This application suggests that an effect greater than that of the original intervention applied can potentially be observed in a cell that received an intervention and also had spillover from another intervention cell. It is also to be expected that any change in the estimated effects would not be observed in spatial settings where the treatment allocation does not satisfy block stratification conditions (where no intervention cells are rook neighbors with other intervention cells). Figure 2 provides a comparative representation of how these two methods permit the application of spillover effects on clusters based on their treatment assignment. In all scenarios of this simulation study, the application of any spillover effect is simply a binary indication of whether a cluster receives a spillover effect or not. Even if spillover exposure could potentially be directed from multiple neighboring clusters, the modeling in this study does not consider any additive spillover effects.

2.4 Simple Random vs. Block Stratified Sampling

In this study, two different treatments are considered: intervention & control. Assignment of these treatments is considered under two sampling methods: Simple Random Sampling (SRS) & Block Stratified Sampling (BSS). Random sampling treatment assignment considers all possible combinations of intervention & control arrangements on a specified spatial grid and provides equal probability of assignment for any given combination of intervention and control. For example, a 2 row by 3 column spatial grid has 6 total cells. Consideration of a 1:1 ratio between intervention and control treatments provides that three cells are assigned as intervention and three cells are assigned as control. Therefore, there are ${}_6C_3 = 20$ different combinations of treatments for this specific spatial region.

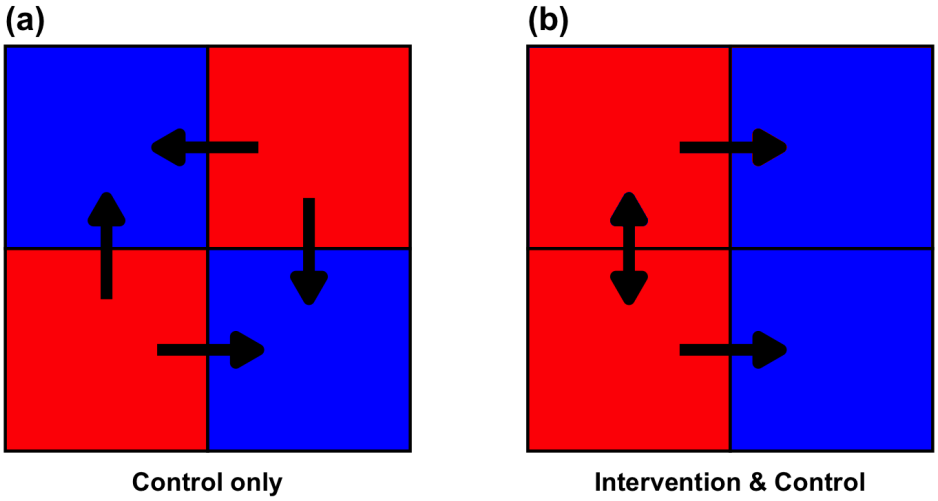


Figure 2: (a) TrtNoSpill: Spillover arrows flow *outward* from intervention cells (red) into neighboring control cells (blue) only; no spillover enters other intervention cells. (b) TrtSpill: Spillover arrows flow from intervention cells (red) into *both* neighboring control cells (blue) and neighboring intervention cells (red).

Block stratified sampling ensures a balanced number of intervention and control cells are selected, with the added condition that no intervention or control cells share a common edge boundary. This approach is analogous to geographically stratified randomization in standard CRTs, where spatial proximity is used to define strata rather than demographic or clinical covariates[35][36][3]. The added spatial isolation condition, that no two same-treatment clusters may share a Rook edge, is a novel constraint introduced in this work to directly minimize cross-cluster spillover at the design stage. An exception to this rule occurs in spatial setups where both dimensions are odd. In this case, the count of intervention and control cells must be out of balance and we assert that the control class will always be in the majority. In these instances, we relax the common edge boundary restriction for control cells and only require that intervention cells not share a common edge boundary in order to satisfy the block stratification condition.

We now consider a series of spatial grid setups to address efficient estimation of the intervention effect. These spatial grids have dimensions denoted as (row $\times$ column) and we seek to apply a balanced distribution between intervention and control cells where possible. When this is not possible, we defer to the control group being the majority class as mentioned previously.

2.4.1 2×4 Grid

One representative spatial setting considered is grid of 8 individual cells laid out in an arrangement of 2 rows and 4 columns. For this grid setup, a balanced set of intervention and control cells can be assigned where 4 cells are intervention regions and 4 cells are control regions. For a grid of 8 total cells and 4 cells of each treatment type, there are ${}_8C_4 = 70$ unique arrangements of treatment and control assignments. Figure 3 (a) displays one of these combinations that satisfies the block stratified condition where intervention and control cells do not have any rook neighbors of the same treatment type. For this spatial setting, there are 2 (out of 70) combinations that satisfy the block stratification condition. Figure 3 (b) displays an example of the 2×4 grid spatial setting with a randomly sampled arrangement of intervention and control cells. Note that both treatment types have cells that are rook neighbors with the same treatment. These spatial grids are constructed with Simple Features for R package[37][30].

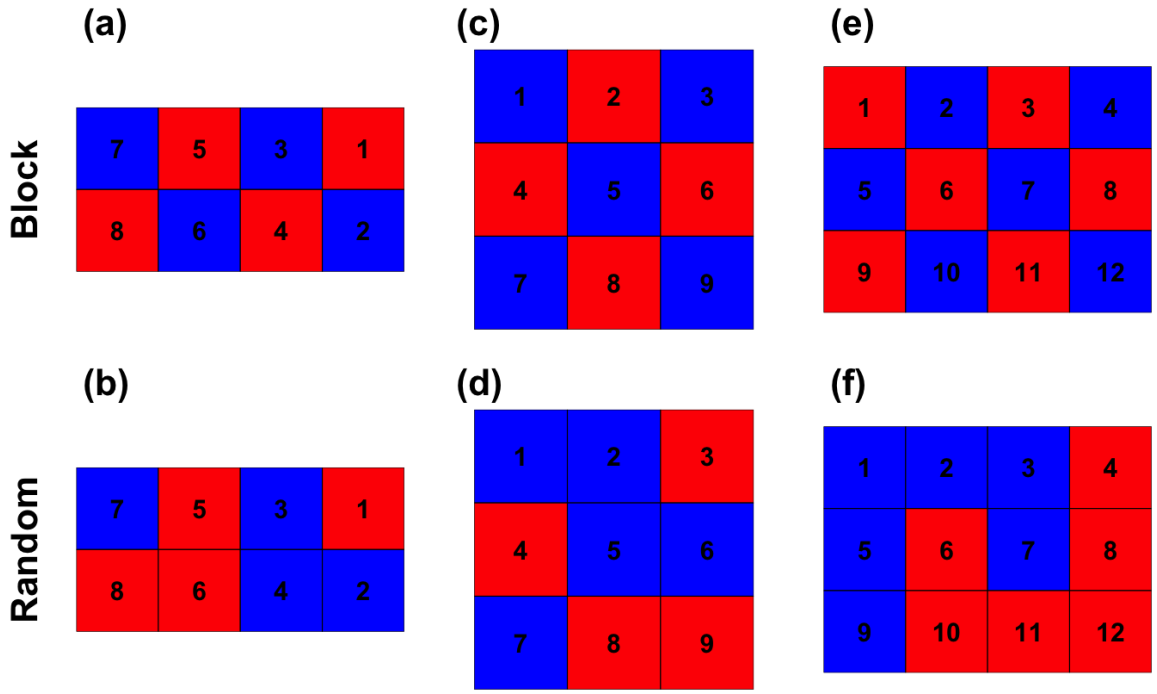


Figure 3: 2×4 (left), 3×3 (middle), and 3×4 (right) spatial grids of intervention (red) and control (blue) clusters assigned via Block Stratified Sampling (top row) and Random Sampling (bottom row)

2.4.2 3×3 Grid

The 3×3 symmetric grid setting of 9 cells allows for ${}_9C_4 = {}_9C_5 = 126$ unique combinations of intervention and control cells where 5 cells are allocated as control and 4 cells are treatment regions. The majority class of 5 cells is the control group for the sake of hypothetical resource management. Figure 3 (c) illustrates one combination of treatment and control cells that completely satisfies the block stratification conditions. Additional cluster combinations satisfy a relaxed condition of block stratification where control cells are allowed to have other control cells as rook neighbors. This is possible in spatial grid settings where both dimensions (row, column) are odd in length. There is one combination for the 3×3 grid setting that satisfies the strongest block stratification condition and 5 more combinations that satisfy the relaxed block stratification condition for a total of 6. An example of random sampling treatment assignment that does not satisfy block stratification conditions is provided in Figure 3 (d).

2.4.3 3×4 Grid

The final spatial setting considered was a 3 row by 4 column (3×4) grid of 12 total cells. This grid, with a balanced allocation of 6 intervention cells and 6 control cells allows for ${}_{12}C_6 = 924$ unique arrangements of the two treatments in the spatial region. For this particular grid size, 2 of the 924 treatment combinations satisfy the block stratification condition and Figure 3 (e) presents one of these, with a "checkerboard" design. The remaining treatment combinations are achieved via random sampling of a balanced number of intervention and control cells. Figure 3 (f) illustrates another randomly sampled arrangement of the treatments onto the spatial area where control and intervention cells each have rook neighbors of the same treatment type.

2.5 Spatial Autoregressive (SAR) Model

Outcomes in the presence of spatial dependence and spillover effects can be considered via modeling with a spatial autoregressive model[38][27][39]. For the design considered in this study, the spatial autoregressive model for the cluster-level outcome has the form:

$$y_i = \alpha + \beta x_i + \psi z_i + \rho \sum_{j \in \mathcal{N}(i)} w_{ij} y_j + \varepsilon_i, \tag{1}$$

where y_i is the cluster-level outcome for cluster i (the modeled quantity); α is the baseline effect; β is the intervention effect; $x_i \in \{0, 1\}$ is the cluster-level treatment indicator (1 if cluster i receives the intervention, 0 otherwise); ψ is the scalar spillover parameter (in this study, spillover is determined by binary cluster adjacency, with the indicator z_i capturing which clusters are exposed, so ψ is a scalar multiplier rather than a function of distance); $z_i \in \{0, 1\}$ is the cluster-level spillover indicator (1 if cluster i is a Rook neighbor of at least one treated cluster and the spillover type permits it; 0 otherwise); $\mathcal{N}(i)$ denotes the set of clusters that share a Rook edge with cluster i (i.e., its immediate horizontal and vertical neighbors); w_{ij} are elements of the row-normalized spatial weights matrix $\mathbf{W}$, equal to $1/|\mathcal{N}(i)|$ for each Rook neighbor j of cluster i and 0 otherwise (full construction in Appendix A); ρ is the spatial autocorrelation coefficient; and ε_i is the i.i.d. error term for cluster i. In matrix form,

$$\mathbf{y} = \boldsymbol{\alpha} + \mathbf{X}\beta + \mathbf{Z}\psi + \rho\mathbf{W}\mathbf{y} + \boldsymbol{\varepsilon}.$$

Within any given cluster, all subjects share the same intervention status (x_i), spillover exposure (z_i), and spatial neighborhood structure (w_{ij}); the only within-cluster variation in outcomes is through an independent (i.i.d.) random error. The outcome y_i is therefore modeled at the cluster level, reflecting the shared structural signal across all subjects in cluster i. This model specification does not parameterize intra-cluster correlation (ICC). Rather, the cluster-level outcome y_i captures the structural and design signal while individual variation is treated as independent noise.

It is important to clarify the conceptual distinction between two spatial quantities in this model: the spatial autocorrelation parameter ρ and the spillover effect ψ. The parameter ρ is a *nuisance* parameter reflecting background spatial correlation in outcomes: nearby clusters tend to produce similar outcomes due to shared environmental, demographic, or social factors, independent of any intervention. In contrast, the spillover effect ψ is a *structural, design-level* parameter, specifically the coefficient on the spillover indicator z_i, representing the indirect (spillover) effect that an intervention in one cluster exerts on outcomes in neighboring clusters. While both quantities involve spatial proximity, ρ is an intrinsic property of the data-generating process that must be accounted for in order to obtain valid inferences, whereas ψ is of direct scientific interest and often a primary estimand alongside the intervention effect β. Failure to distinguish these two quantities, or to include both in the model, can produce confounded estimates of the intervention effect. In particular, when spatial dependence is high, failing to include ρ in

the model may cause the spillover estimate $\hat{\psi}$ to absorb background correlation, overstating the apparent indirect effect of the intervention.

The quantities β, ψ, and ρ are *structural coefficients* of the spatial autoregressive model rather than marginal potential-outcome contrasts. This distinction matters because the SAR specification in Equation (1) places the response on both sides of the equation, so the model's reduced (equilibrium) form is

$$\mathbf{y} = (\mathbf{I} - \rho\mathbf{W})^{-1} (\boldsymbol{\alpha} + \mathbf{X}\beta + \mathbf{Z}\psi + \boldsymbol{\varepsilon}), \tag{2}$$

where $\mathbf{X}$ and $\mathbf{Z}$ are the cluster-level intervention and spillover indicator vectors (with entries x_i and z_i), $\boldsymbol{\alpha}$ is the common baseline, and $\boldsymbol{\varepsilon}$ is the error vector. A change in any cluster's treatment status therefore propagates to every cluster through the spatial multiplier $(\mathbf{I} - \rho\mathbf{W})^{-1}$. As a consequence, a potential-outcome indirect effect such as $E[Y(0, \mathbf{1})] - E[Y(0, \mathbf{0})]$ (the contrast between every other cluster being treated versus untreated while a focal cluster is held at control) is a *derived* quantity that depends jointly on β, ψ, ρ, and $\mathbf{W}$, and equals no single coefficient in the model. In particular, when $\rho \neq 0$ even "control-only" spillover induces propagation among treated clusters through the multiplier, so ψ should be read as the structural spillover coefficient on z_i rather than as the marginal indirect effect itself. Accordingly, the simulation study that follows evaluates how well each design recovers the *structural* intervention coefficient β used in the data-generating process, and neither β nor ψ is equated with a marginal potential-outcome contrast.

2.6 Spatial Dependence

Spatial dependence, used interchangeably with spatial autocorrelation, refers to the property of spatial data where observations close to each other tend not to be independent of each other. Rather, observations that are close to each other tend to be dependent on each other and influence values of surrounding data. Hence, there is an implied systematic correlation between data at neighboring locations. Spatial correlation can be positive, where similar values tend to be clustered close together, or negative, where dissimilar values are close together. Appropriate consideration of spatial dependence is necessary for performing valid statistical inference as well as for accurate estimation of effects that are modeled. A pair of common methods for assessing spatial autocorrelation are Moran's I [40][41] and Geary's C [42][38]. Moran's I standardizes the spatial autocovariance by the variance of the data and

Geary's C utilizes the sum of the squared differences between pairs of data as a measure of co-variation. These test statistics are reliant on the spatial weights matrix. Both can be computed in R via the **spdep** package[29].

The spatial weights matrix $\mathbf{W}$ encodes the strength of spatial connections between clusters; its elements w_{ij} can be constructed as indicator-based (Rook contiguity) or distance-based weights, both of which are row-normalized to ensure comparability across clusters with differing numbers of neighbors. In this study, row-normalized indicator (Rook contiguity) weights are used throughout. The spatial dependence term of the SAR model is therefore

$$\rho \mathbf{W}\mathbf{y} = \rho \sum_{j \in \mathcal{N}(i)} w_{ij} y_j, \tag{3}$$

where j indexes clusters and $w_{ij} = 0$ whenever clusters i and j are not Rook neighbors, so the sum reduces naturally to neighboring clusters only. Full definitions of indicator-based and distance-based weight construction, including row-normalization formulas, are provided in Appendix A.

2.7 Estimation for SAR Model

Maximum likelihood estimation is used to estimate the cluster-level spatial lag model in Equation (1)[43][44][45]. The full likelihood and log-likelihood expressions, along with details of the MLE derivation, are provided in Appendix B. In practice, estimates are obtained via the `lagsarlm()` function from the **spdep** R library[28].

3 Simulation Study

A simulation study was carried out to consider the contrast of block stratified and random treatment assignment sampling methods with consideration for spillover effects, spatial dependence, and restrictions on spillover application between intervention and control regions. The results of this study are detailed in the remainder of this section.

3.1 Simulation Scenarios & Response

We carry out a simulation study to explore how estimation error of the intervention effect in a Spatial CRT varies across a range of spillover effect sizes, the presence or absence of spatial dependence, varying

restrictions on clusters where spillover is permitted to occur, and the method of sampling for treatment assignment. Each simulation scenario is composed of a set of these parameters. The baseline effect, α, is always assigned as 0.2. The intervention effect, β, is always assigned as 1.0. The spillover effect, ψ, has possible values 0.5, 0.6, 0.7, and 0.8. The spatial dependence parameter, ρ, is set as either 0.00 or 0.01. The sampling design for each scenario is either Block Stratified Sampling (BSS) or Simple Random Sampling (SRS). Finally, the spillover application type is (1) Control Clusters Only or (2) Control & Intervention Clusters. Each simulation setting is replicated for each of the spatial grid dimensions discussed previously (2×4, 3×3, and 3×4).

Given that the form of the spatial lag model in Section 2.5 contains the response on both sides of the expression, outcomes are generated from the reduced (equilibrium) form derived in Equation (2)[27][46], with $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I})$ and $\sigma^2 = 0.01$, and simulation parameter values $\alpha = 0.2$, $\beta = 1.0$, $\psi \in \{0.5, 0.6, 0.7, 0.8\}$, and $\rho \in \{0.00, 0.01\}$ as specified above.

Block stratified sampling is itself the treatment allocation mechanism: an investigator adopting BSS deliberately selects one of the spatially isolated allocations that satisfy the block stratification condition, rather than drawing an allocation at random from the full set of balanced assignments. Because only a small number of allocations satisfy this condition (2 for the 2×4 grid, 6 for the 3×3 grid, and 2 for the 3×4 grid), BSS does not induce a usable randomization distribution, and we make no claim that it supports permutation-based or randomization-based inference. This near-determinism is a deliberate property of the design rather than a defect: by restricting allocation to spatially favorable arrangements, BSS removes the high-error allocations that simple random sampling can draw by chance, bounding an investigator's worst-case estimation error across the feasible allocations for a given grid.

Frequentist inference for the intervention effect does not, however, rely on a randomization distribution over allocations. For any single realized allocation, including a BSS allocation, the estimator $\hat{\beta}$ has a well-defined sampling distribution arising from the stochastic outcome process: model-based inference via SAR maximum likelihood estimation (the `lagsarlm()` R function) conditions on the realized allocation and derives this distribution from the random error term, independent of how many alternative allocations exist. Model-based SAR inference is therefore the appropriate framework for practitioners implementing either SRS or BSS in an applied study, and the small size of the BSS allocation set does not invalidate it. The replication-to-replication variation in our simulation is correspondingly outcome-level

Monte Carlo variation for a fixed allocation, not variation induced by re-randomizing the allocation. Within this framing, the relevant question is not how many allocations a design admits but whether the principled fixed allocation it selects yields a low-error estimator. This study uses exhaustive enumeration of all possible treatment allocations, feasible here given the small grids (70, 126, and 924 total combinations, respectively), precisely to characterize that full design space and identify which sampling strategy produces more favorable bias–variance properties when the true data-generating model (SAR with spillover) is correctly specified.

3.2 Simulation Procedure

The algorithm implemented to perform the simulations for this study is illustrated in the following section. The simulation is performed over all potential cluster treatment combinations (intervention/control) for each of the 2×4, 3×3, and 3×4 cluster spatial grid geometries. The simulation procedure is as follows:

For each spatial grid geometry and simulation scenario, the complete set of possible treatment combinations was enumerated exhaustively (70 for the 2×4 grid, 126 for the 3×3 grid, and 924 for the 3×4 grid). For each combination, $N = 50$ simulation replications were performed for the 2×4 and 3×3 grids and $N = 10$ for the 3×4 grid, with 20 subjects uniformly sampled per cluster in each replication. Outcomes were generated from the SAR closed-form data-generating process (Equation 2), and parameters $(\alpha, \beta, \psi, \rho)$ were estimated via `lagsarlm()` for each replication. BSS-valid combinations were identified and tracked separately (2 for the 2×4 grid, 6 for 3×3, and 2 for 3×4). The complete step-by-step algorithm, including BSS identification criteria and aggregation procedures, is provided in Appendix C.

The structural parameters $(\alpha, \beta, \psi, \rho)$ were estimated jointly for each replication via SAR maximum likelihood, with both the intervention indicator x_i and the spillover indicator z_i included alongside the spatial lag term (Equation 1). In scenarios where spillover propagates only to adjacent control clusters, however, the spillover indicator z_i is determined by adjacency to intervention clusters and can become highly correlated with, or in certain allocation configurations exactly collinear with, the intervention indicator x_i. When such collinearity is present, the jointly estimated $\hat{\beta}$ may absorb part of the spillover signal, producing a systematic negative bias approximately equal to $-\psi$. This mechanism underlies the $\hat{\beta}$ bias pattern visible in the control-clusters-only results (Tables 1–3).

Figure 4 presents an example of the 8 cluster 2×4 spatial grid object. 160 subjects (20 subjects for each cluster) with uniformly distributed coordinates are sampled over the area of the spatial object and colored according to their cluster membership. Subject counts are not necessarily identical between any pair of clusters as the sample of subjects is considered over the whole space. Points in any cluster on the grid are subject to the selected treatment (intervention/control) and experience the same spillover exposure, either all exposed or all unexposed, as every other point in that cluster. The individual within-cluster coordinates are used only to assign subjects to clusters and to visualize the sampled grid; beyond determining cluster membership, they are not consequential to the SAR fit. All cluster-level quantities $(x_i,\ z_i,\ w_{ij})$ are shared by every subject in the cluster, so the fitted model depends on each subject only through cluster membership. Within-cluster variation in individual outcomes contributes to the error-variance estimate but carries no information about β, ψ, or ρ. Treating spillover or treatment as a function of a specific individual's location within a cluster is a natural extension left to future work.

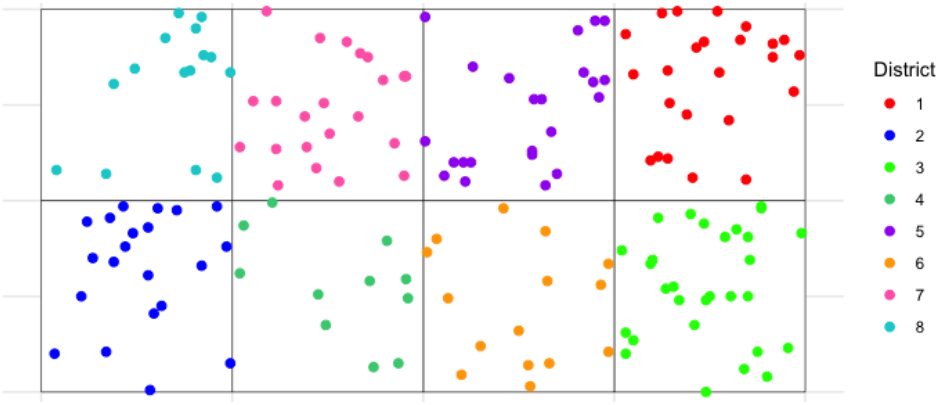


Figure 4: Example of simulated observation coordinates (n=160) on a 2×4 grid of clusters

3.3 Simulation Evaluation Metrics

Estimates for the baseline effect (α), intervention effect (β), spillover effect (ψ), and spatial correlation parameter (ρ) were collected for each iteration of the simulation procedure for every treatment assignment combination. Considering the initial fixed simulation parameters, the following metrics are computed for estimation of the intervention effect for the evaluation of each set of simulation conditions.

Bias: a measure of how different the estimated parameter value is from the true parameter value. Small bias means that the parameter estimate is quite close to the true value.

Variance: a measure of the spread of a set of estimates from multiple simulation iterations. Smaller variance indicates that the estimates are tightly grouped together and large variance means that the estimates are spread out.

Mean Squared Error (MSE): measures the average squared difference between each estimate and the true value of a parameter. Mean Squared Error combines variance to understand the consistency of a set of estimates as well as bias to understand any trend in deviation from the true value into a measure of estimation accuracy.

Since MSE = $\text{Bias}^2 + \text{Variance}$, differences in MSE between SRS and BSS can be attributed to either systematic bias or sampling variability across allocations. This decomposition is visually reflected in the grouped bar figures throughout the Results section (Figures 5–7), where the relative contribution of bias-driven versus variance-driven error is apparent from differences in bar height across sampling methods and conditions.

Mean Squared Error is the primary metric of consideration to understand the efficient estimation of the intervention effect under all simulation settings. The following subsections detail the findings from the simulation study, focusing on the Mean Squared Error (MSE), standard deviation (SD), and bias of the intervention effect (β) estimates under various conditions (simulation scenarios). The analysis compares Simple Random Sampling (SRS) and Block Stratified Sampling (BSS) across different spatial grid configurations, spillover effect magnitudes, spatial dependence levels, and spillover application types.

3.4 Intervention Effect Estimates (2×4 grid)

For the 2×4 spatial grid, the simulation results (Table 1, Figure 5) reveal a risk-management trade-off between SRS and BSS rather than a uniform advantage for either method.

Under the Control Clusters Only (TrtNoSpill) spillover condition, performance depends on the level of spatial dependence. When spatial dependence is present ($\rho = 0.01$), BSS substantially outperforms SRS (e.g., BSS MSE = 0.165 vs. SRS MSE = 0.304 at $\psi = 0.5$). However, when spatial dependence is absent ($\rho = 0$), SRS achieves lower average MSE (e.g., SRS MSE = 0.173 vs. BSS MSE = 0.262 at $\psi = 0.5$). Crucially, the standard deviation (SD) of BSS estimates is consistently far smaller than that

of SRS across all conditions under TrtNoSpill. This reflects the fact that the two BSS allocations for this grid are structurally similar, yielding reliably consistent estimation error, while the 70 SRS allocations span a wide range of spatial configurations, some well-balanced and some poorly concentrated, producing highly variable MSE.

Under the Intervention & Control Clusters (TrtSpill) spillover condition, SRS achieves lower average MSE in every scenario (e.g., SRS MSE = 0.008 vs. BSS MSE = 0.262 at $\psi = 0.5$, $\rho = 0$). However, the SD of SRS estimates is substantially larger, meaning that while the average SRS allocation performs better, some randomly drawn allocations carry considerable estimation risk. BSS delivers stable, though higher, average MSE with near-zero SD, ensuring that practitioners never encounter a catastrophically poor allocation.

The presence of spatial dependence ($\rho = 0.01$) ~~consistently increased MSE for both methods~~ increased the MSE of SRS relative to the no-dependence setting in every scenario, a pattern that persists across all grid sizes. For BSS on this grid, spatial dependence lowered MSE (e.g., 0.165 vs. 0.262 at $\psi = 0.5$). The central message from the 2×4 results is that BSS bounds the worst-case outcome at the cost of a somewhat higher average MSE in certain spillover regimes, whereas SRS offers the possibility of a lower average MSE but with substantially greater risk of a high-MSE realization.

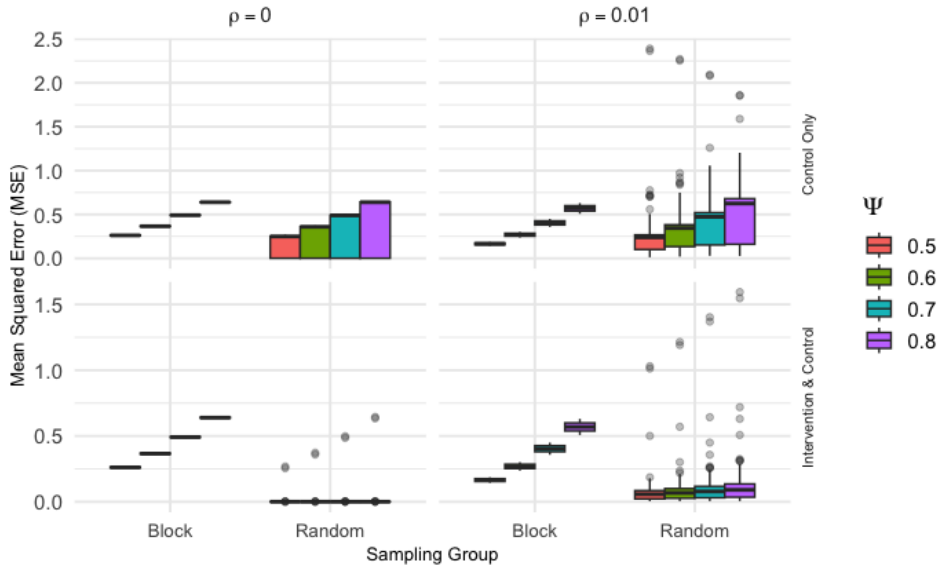


Figure 5: Intervention effect MSE for 2×4 cluster grid grouped by spillover effect magnitude, spatial dependence, sampling design, and spillover application type

Spillover Application	Spatial Dependence (ρ)	Spillover Effect (ψ)	Simple Random Sampling			Block Stratified Sampling		
			MSE	SD ($\times 10$)	Bias	MSE	SD ($\times 10$)	Bias
Control Only Spillover	Yes (0.01)	0.5	0.304	4.053	-0.270	0.165	0.294	-0.349
		0.6	0.382	4.062	-0.342	0.269	0.462	-0.471
		0.7	0.471	4.143	-0.414	0.404	0.653	-0.594
		0.8	0.573	4.390	-0.486	0.569	0.867	-0.716
	No (0)	0.5	0.173	1.170	-0.343	0.262	0.095	-0.496
		0.6	0.248	1.683	-0.411	0.366	0.100	-0.597
		0.7	0.338	2.291	-0.480	0.492	0.095	-0.697
		0.8	0.441	2.994	-0.548	0.639	0.081	-0.798
Intervention & Control Spillover	Yes (0.01)	0.5	0.095	1.762	0.031	0.165	0.294	-0.349
		0.6	0.115	2.082	0.030	0.269	0.462	-0.471
		0.7	0.138	2.423	0.029	0.404	0.653	-0.594
		0.8	0.162	2.787	0.027	0.569	0.867	-0.716
	No (0)	0.5	0.008	0.438	-0.014	0.262	0.095	-0.496
		0.6	0.011	0.614	-0.017	0.366	0.100	-0.597
		0.7	0.015	0.825	-0.020	0.492	0.095	-0.697
		0.8	0.019	1.072	-0.023	0.639	0.081	-0.798

Table 1: SRS vs. BSS: Intervention Effect Error Estimates - 2×4 Spatial Grid Setting

3.5 Intervention Effect Estimates (3×3 grid)

The results for the 3×3 spatial grid are summarized in Table 2 and Figure 6.

For the Control Clusters Only (TrtNoSpill) spillover application, BSS ~~consistently yielded significantly~~ yielded lower MSE, SD, and bias ~~compared to SRS . For instance, with~~ than SRS in every scenario. With $\rho = 0$, BSS MSE ~~was minimal (e.g., 0.001 for $\psi = 0.5$),~~ ranged from 0.042 to 0.106, while SRS MSE ranged from ~~0.092 to 0.235. This pattern held true with~~ 0.114 to 0.291. With spatial dependence ($\rho = 0.01$), ~~where BSS MSE remained very low (e.g., 0.004 for $\psi = 0.5$), contrasting with SRSMSE values ranging from 0.259~~ BSS MSE ranged from 0.147 to ~~0.372. SRS estimates consistently~~ 0.240, against 0.274 to 0.455 for SRS. Both designs showed negative bias ~~, increasing~~ that grew in magnitude with ψ, ~~while BSS bias was negligible.~~ and the bias was smaller for BSS (e.g., -0.078 vs. -0.226 at $\rho = 0$, $\psi = 0.5$).

In the Intervention & Control Clusters (TrtSpill) spillover scenario, ~~BSS continued to demonstrate superior performance with substantially lower MSE, SD, and bias compared to SRS~~SRS achieved lower average MSE than BSS in every scenario. For $\rho = 0$, ~~BSS MSE was consistently around 0.0004, whereas~~ SRS MSE ranged from 0.002 to 0.005, whereas BSS MSE ranged from 0.042 to 0.106. With $\rho = 0.01$, ~~BSS MSE was also consistently very low (e.g., 0.039 for $\psi = 0.5$), while~~ SRS MSE ranged from 0.103

to ~~0.168.~~ 0.167 and BSS MSE from 0.147 to 0.240. Unlike the 2×4 and 3×4 grids, where the BSS allocations are the two checkerboards, the six BSS allocations on this grid differ in structure, so their MSE also varies across allocations. Under TrtSpill, the SD of BSS was larger than that of SRS.

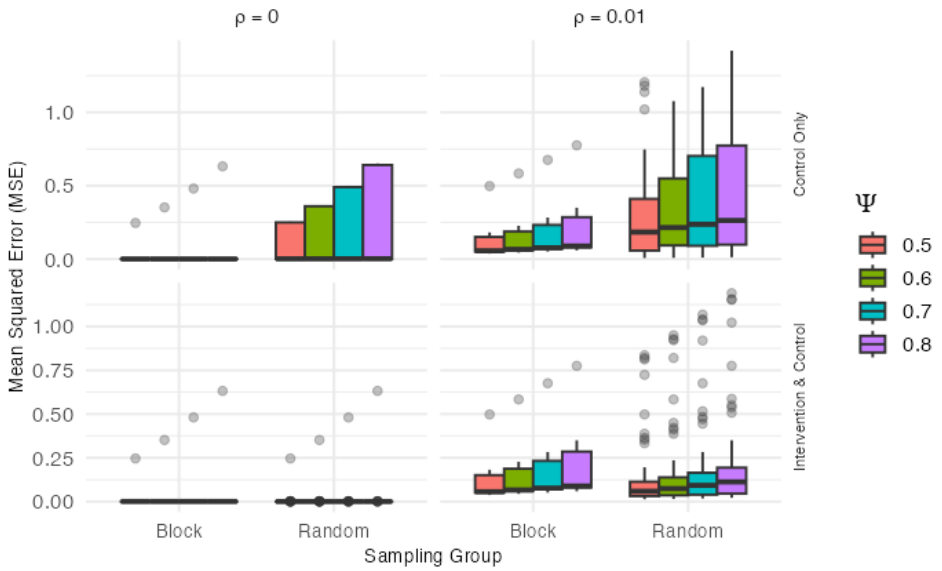


Figure 6: Intervention effect MSE for 3 × 3 cluster grid grouped by spillover effect magnitude, spatial dependence, sampling design, and spillover application type

Spillover Application	Spatial Dependence (ρ)	Spillover Effect (ψ)	Simple Random Sampling			Block Stratified Sampling		
			MSE	SD (×10)	Bias	MSE	SD (×10)	Bias
Control Only Spillover	Yes (0.01)	0.5	0.274	2.631	-0.089	0.147	1.799	-0.054
		0.6	0.330	2.964	-0.135	0.174	2.115	-0.064
		0.7	0.390	3.472	-0.181	0.205	2.460	-0.073
		0.8	0.455	4.140	-0.227	0.240	2.839	-0.083
	No (0)	0.5	0.114	1.247	-0.226	0.042	1.002	-0.078
		0.6	0.164	1.798	-0.271	0.059	1.434	-0.096
		0.7	0.223	2.449	-0.316	0.081	1.959	-0.113
		0.8	0.291	3.203	-0.362	0.106	2.577	-0.130
Intervention & Control Spillover	Yes (0.01)	0.5	0.103	1.480	0.071	0.147	1.799	-0.054
		0.6	0.122	1.688	0.082	0.174	2.115	-0.064
		0.7	0.143	1.905	0.092	0.205	2.460	-0.073
		0.8	0.167	2.133	0.102	0.240	2.839	-0.083
	No (0)	0.5	0.002	0.219	-0.004	0.042	1.002	-0.078
		0.6	0.003	0.313	-0.004	0.059	1.434	-0.096
		0.7	0.004	0.428	-0.005	0.081	1.959	-0.113
		0.8	0.005	0.562	-0.006	0.106	2.577	-0.130

Table 2: SRS vs. BSS: Intervention Effect Error Estimates - 3 × 3 Spatial Grid Setting

3.6 Intervention Effect Estimates (3×4 grid)

Table 3 and Figure 7 present the intervention effect estimates for the 3×4 spatial grid. These results ~~reveal both the typical strengths of BSS and a notable exception that illuminates the limits of the checkerboard design~~follow the same pattern as the 2×4 grid in every scenario.

Under the Control Clusters Only (TrtNoSpill) spillover condition, ~~BSS dramatically outperforms SRS across all combinations of ρ and ψ. With no~~ performance again depends on the level of spatial dependence. With spatial dependence (~~$\rho = 0$~~$\rho = 0.01$), BSS ~~MSE is an order of magnitude smaller~~ achieves lower MSE than SRS (e.g., BSS MSE ~~≈ 0.0004~~ $= 0.133$ vs. SRS MSE ~~$= 0.034$~~ $= 0.244$ at $\psi = 0.5$). With no spatial dependence (~~$\rho = 0.01$), BSS retains a substantial advantage~~ $\rho = 0$), SRS achieves lower average MSE (e.g., ~~BSS MSE ≈ 0.035 vs.~~ SRS MSE ~~$= 0.120$~~ $= 0.133$ vs. BSS MSE $= 0.226$ at $\psi = 0.5$). ~~BSS SD is near zero throughout, while SRS SD is substantially larger, confirming that~~ The SD of BSS is smaller than that of SRS throughout, because the two valid BSS checkerboard allocations for this grid ~~provide stable and consistently low-error estimates whenever spillover is confined to control clusters~~give similar estimation error.

Under the Intervention & Control Clusters (TrtSpill) spillover condition, ~~results are more nuanced. With no spatial dependence ($\rho = 0$), BSS achieves lower MSE than SRS~~ SRS achieves lower average MSE in every scenario (e.g., ~~BSS MSE ≈ 0.0003 vs.~~ SRS MSE ~~$= 0.001$~~ $= 0.112$ vs. BSS MSE $= 0.133$ at $\psi = 0.5$~~). However, when spatial dependence is present (~~, $\rho = 0.01$~~), BSS MSE (≈ 0.223 at $\psi = 0.5$)substantially exceeds SRS MSE ($= 0.112$). This inversion occurs because the strict checkerboard structure of the 3×4 BSS allocation ensures that every intervention cluster is surrounded entirely by control neighbors eligible for spillover ; under TrtSpill, this configuration amplifies the systematic bias in $\hat{\beta}$, a "checkerboard paradox" discussed further in Section 5~~, and 0.001 vs. 0.226 at $\rho = 0$). In both checkerboard allocations, every control cluster borders an intervention cluster and no intervention cluster borders another, so the spillover indicator equals $1 - x_i$ and the two spillover types produce identical BSS results. β and ψ are then not separately identifiable, and $\hat{\beta}$ absorbs the spillover effect, with bias near $-\psi$ when $\rho = 0$ (e.g., -0.463 at $\psi = 0.5$; Section 5.2).

The 3×4 results reinforce the broader conclusion: BSS ~~reliably~~ bounds worst-case estimation error ~~under TrtNoSpill, while the TrtSpill regime~~ and has its advantage under TrtNoSpill with spatial dependence~~represents the primary scenario in which a practitioner should carefully weigh the risk-management~~

~~benefits of BSS against its potential for amplified bias~~, while SRS offers lower average MSE when spillover reaches both treatment arms.

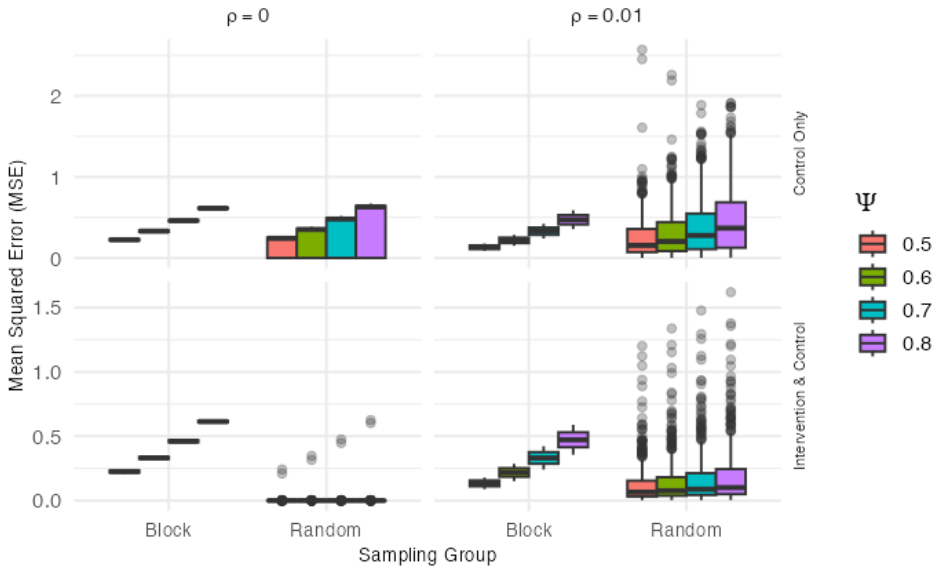


Figure 7: Intervention effect MSE for 3×4 cluster grid grouped by spillover effect magnitude, spatial dependence, sampling design, and spillover application type

Spillover Application	**Spatial Dependence (ρ)**	**Spillover Effect (ψ)**	**Simple Random Sampling**			**Block Stratified Sampling**		
			MSE	SD ($\times$100)	Bias	MSE ($\times$10)	SD ($\times$100)	Bias
Control Only Spillover	Yes (0.01)	0.5	0.244	24.422	-0.190	1.332	6.416	-0.273
		0.6	0.311	28.926	-0.242	2.180	9.471	-0.391
		0.7	0.390	34.937	-0.292	3.309	12.806	-0.509
		0.8	0.480	42.507	-0.341	4.721	16.429	-0.627
	No (0)	0.5	0.133	12.497	-0.264	2.255	2.274	-0.463
		0.6	0.191	18.005	-0.317	3.315	2.196	-0.570
		0.7	0.260	24.518	-0.370	4.609	1.991	-0.676
		0.8	0.339	32.042	-0.423	6.137	1.659	-0.782
Intervention & Control Spillover	Yes (0.01)	0.5	0.112	12.873	0.022	1.332	6.416	-0.273
		0.6	0.131	14.762	0.021	2.180	9.471	-0.391
		0.7	0.151	16.817	0.020	3.309	12.806	-0.509
		0.8	0.173	19.046	0.018	4.721	16.429	-0.627
	No (0)	0.5	0.001	1.051	-0.001	2.255	2.274	-0.463
		0.6	0.001	1.542	-0.001	3.315	2.196	-0.570
		0.7	0.001	2.143	-0.002	4.609	1.991	-0.676
		0.8	0.002	2.853	-0.002	6.137	1.659	-0.782

Table 3: SRS vs. BSS: Intervention Effect Error Estimates - 3×4 Spatial Grid Setting

4 Application

The findings from our simulation study provide relevant insights for investigators designing CRTs in real-world applications where spatial dependence and spillover effects are present and must be accounted for. In this case, a pair of studies in development that we consider involve the North Carolina Department of Adult Correction. The structure of correctional systems, particularly community supervision, often involves geographically clustered units where staff from different clusters may interact and regional similarities can influence outcomes. The investigators leading these studies desire recommendations for treatment assignment methodology regarding the spatially clustered nature of their design.

4.1 Background of Probation in North Carolina

The North Carolina Department of Adult Correction Division of Community Supervision (NC DCS) provides a clear example of a spatially structured organization, making it an ideal context for applying our research. The NC DCS is organized into four geographic Divisions numbered 1 through 4 from East to West, which are subdivided into 30 Districts that make up the entirety of the state. Each district is composed of one or more counties. If there is more than one county in the district, the counties are adjacent, but some districts are composed of only a singular county. In these districts, Probation and Parole Officers (PPOs) supervise individuals at the county level and are themselves managed by Chief Probation/Parole Officers (CPPOs) and Judicial District Managers (JDMs) who oversee districts. Each division (composed of multiple districts) is managed by a Judicial Division Director (JD Dir) as well. A map of the segmentation of each division and district is presented in Figure 8.

This geographic and administrative structure means that Chiefs and Judicial District Managers (JDMs) often span multiple units and counties, creating a direct channel for spillover effects among regions under their supervision. For instance, if leadership views an intervention in one county favorably, they may implement its components in other counties under their jurisdiction, even if those areas were not assigned the intervention. Furthermore, adjacent counties within the same region often share demographic characteristics, labor market conditions, and similar judicial cultures, as well as the same managed care organizations and treatment providers, leading to spatial dependence in probation outcomes. The choice of treatment assignment sampling strategy in this context is non-trivial when considering potential options of either simple random sampling (SRS) or block-stratified sampling (BSS).

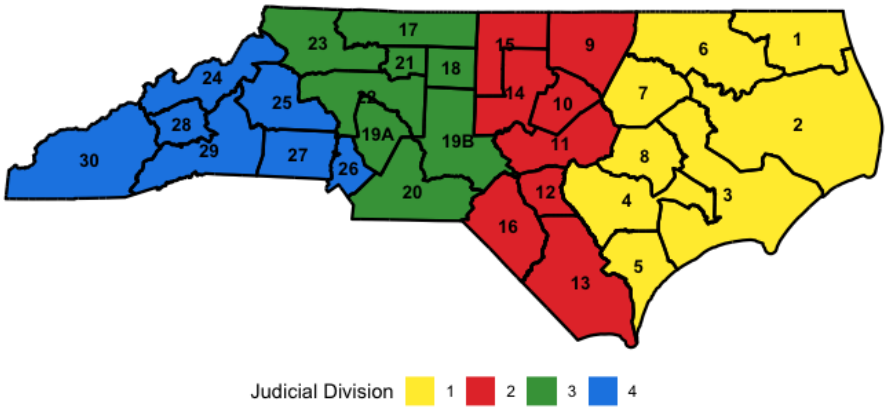


Figure 8: The North Carolina Department of Adult Correction Division of Community Supervision map is segmented into 30 judicial districts that are grouped into 4 judicial divisions.

Selection of the appropriate treatment sampling strategy may yield beneficial downstream outcomes when considering the error of intervention & spillover estimates.

4.2 Case 1: Trauma-Informed and Engaged Supervision

One proposed study involves implementing a trauma-informed supervision model for PPOs managing specialty caseloads like domestic violence, sex offender, or gang affiliated cases. The intervention consists of intensive training and follow-up sessions delivered to cohorts of PPOs and CPPOs. The primary outcomes would be changes in officer knowledge and attitudes, as well as impacts on individuals under their supervision. The key design decision for this study is the unit of randomization. Randomizing at the PPO level is not feasible due to the high likelihood of spillover within a county office. Therefore, a cluster randomized design at the county or district level is necessary. This cluster randomization decision is also supported by agency-related logistics in addition to considerations for spillover. Original plans for a similar study, acknowledging the risk of contamination between adjacent units, considered randomization at the district level specifically to minimize spillover. This scenario directly reflects the conditions modeled in our simulation, where the effectiveness of an intervention in one cluster may influence outcomes in neighboring control clusters where an intervention is not applied.

4.3 Case 2: Testing an implementation strategy for specialty mental health probation

Another study proposal aims to test a strategy for improving the implementation of Specialty Mental Health Probation (SMHP). The intervention involves enhancing collaboration between PPOs and community-based behavioral health service providers through cross-training sessions and joint treatment team meetings. The primary outcome is improved collaboration, with secondary outcomes related to treatment engagement for individuals on probation. This case presents a more complex spatial challenge. In addition to the geographic proximity of probation districts, the behavioral health system in North Carolina is managed by four large Managed Care Organizations (MCOs), each responsible for a different coverage area. Some service provider agencies are large, operating in multiple counties and potentially across different MCO regions. This creates multiple, overlapping layers of potential spillover. Figure 9 presents a map of which counties in the state that each MCO operates. An intervention implemented in one county could be adopted by a provider and informally carried into a neighboring control county, or an MCO could promote a similar initiative, confounding the study's results. This complex spatial linkage underscores the importance of a robust treatment assignment strategy in order to efficiently estimate the impact of the intervention.

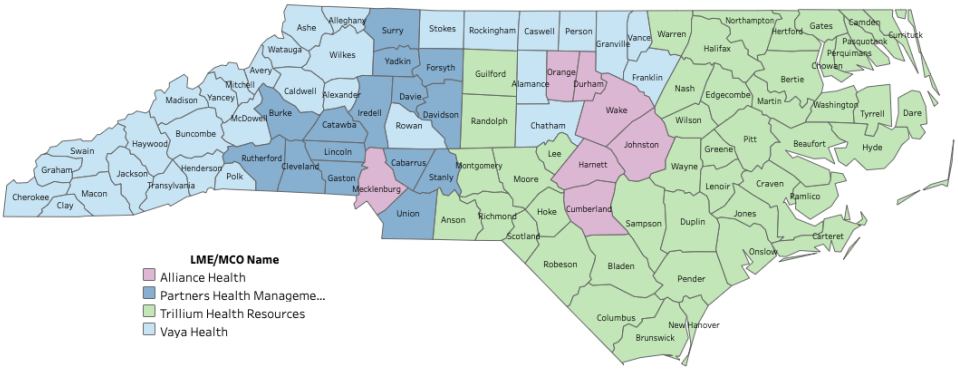


Figure 9: Map of Local Management Entity/Managed Care Organizations (LME/MCOs) for each county in the state of North Carolina. There are 4 LME/MCOs of which one operates in each of the 100 counties of the state.

4.4 Recommendations for treatment assignment

The results of our simulation study indicate a clear trade-off: we previously concluded that Simple Random Sampling (SRS) reduces the average mean squared error (MSE) across many randomizations (intervention/control combinations), while Block Stratified Sampling (BSS) reduces the variance of the error although it is not minimized. Put simply, BSS is less likely to produce a "perfect" assignment by chance but is also robust against a "worst-case" assignment that could render the study results invalid. For both case studies presented, Block Stratified Sampling (BSS) is our recommended approach, subject to an explicit qualification: the recommendation rests on the expectation that spillover in these correctional-supervision settings flows primarily from intervention units into neighboring control units, the regime in which BSS reliably controls estimation variance and bounds worst-case allocation risk. If substantial ~~bidirectional~~ spillover among treated units is instead anticipated~~together with background spatial dependence, the spatial separation enforced by BSS can amplify rather than reduce bias (the "checkerboard paradox,"~~ , SRS achieved lower average MSE in every simulated scenario of that type (Section 5)~~; in~~ . In that regime Simple Random Sampling, or an analysis-stage buffer-zone design (Section 5, Future Considerations), may be preferable. We therefore encourage investigators to state their anticipated spillover mechanism explicitly when justifying the chosen design.

In Case 1 (Trauma-Informed Supervision), a simple random assignment of districts could, by chance, result in a geographic imbalance, concentrating treated districts in one part of the state. This would confound the intervention effect with regional characteristics. A more robust design would be to stratify by the four state divisions and randomize districts to intervention and control conditions within each division with a block stratification strategy. This ensures geographic balance and leverages the finding that BSS minimizes estimation variance, providing a more stable and defensible estimate of the intervention effect.

In Case 2 (SMHP Implementation), the rationale for BSS is even stronger due to the complex potential for spillover. A simple random assignment of counties would ignore the significant structural influence of the MCOs. The most effective design would be to stratify by MCO coverage area, randomizing counties to the implementation strategy (intervention) or control condition within each MCO. This approach directly controls for the significant confounding variable of MCO-level policies and provider networks. However, some challenges with this MCO stratification do persist. For example, it is possible that a county is

selected for the intervention and is supervised by a CPPO or even a judicial district manager that oversees a control group county. Furthermore, multiple service providers may serve a given county in the coverage area of an MCO and it is possible that multiple service providers begin team meetings. Another consideration is that some service providers are very large and have offices around the state and contract with more than one MCO. Based on this, we must consider potential spillover if after participating in a treatment team meeting, a service provider intends to initiate those practices in another part of the state. Given the high risk of contamination, the robustness of BSS in minimizing error variance makes it the most prudent choice for achieving a reliable effect estimate. While SRS might achieve a lower MSE (intervention error estimate) in theory, the risk of a single, poorly distributed randomization in a complex, real-world setting outweighs the potential benefit. Therefore, it is wise to proceed with a sampling strategy that aims to manage estimation error with minimal variation.

A representation of this block stratified sampling strategy where we "block" treatment and control districts together with an effort to isolate districts of a particular intervention status from each other is shown in Figure 10. Note that across the entire map of the state and specifically in Judicial Division 1 (easternmost region) that when possible districts of the same intervention status are isolated from each other. When districts share a border with multiple other districts, this sampling method ensures that treatment districts are isolated from other treatment districts as best as possible. It is important to note that, unlike the idealized rectangular grids examined in the simulation study, real-world jurisdictions have irregular shapes and varying numbers of shared borders. The allocation depicted in Figure 10 therefore represents a *best-effort* block stratification: the assignment minimizes shared borders between same-treatment districts while acknowledging that perfect spatial separation cannot always be achieved in practice. Investigators applying BSS to real-world geographic settings should expect some deviation from the strict no-common-edge criterion and should document the degree to which the chosen allocation approximates the ideal.

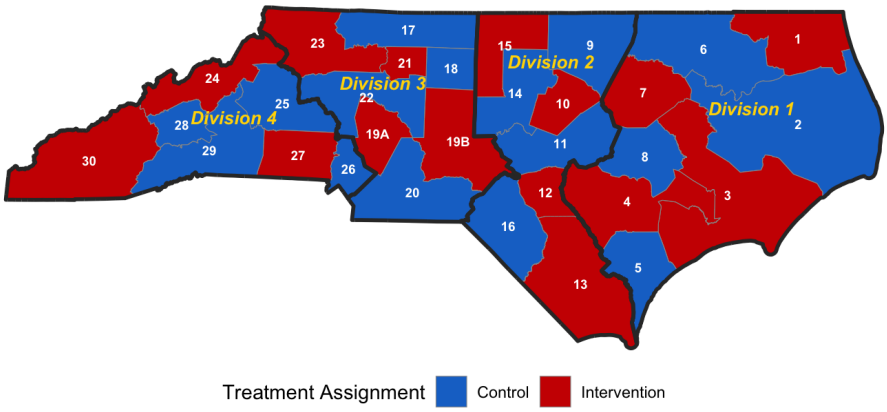


Figure 10: Map of North Carolina with example block stratified sampling intervention assignments for each judicial district under consideration. Districts are assigned to either treatment (red) or control (blue).

5 Discussion

5.1 Relevance of results

This simulation study provides insights into the efficiency of treatment assignment methodologies in spatial cluster randomized trials (CRTs) when considering spillover effects and spatial dependence. A primary contribution of this work is the direct comparison between Simple Random Sampling (SRS) and Block Stratified Sampling (BSS) across various scenarios, offering practical guidance for investigators designing such trials.

The spatial separation principle underlying BSS has antecedents in spatially balanced experimental designs. For example, McCann et al. (2018) demonstrated that spatially stratified allocation in field trials reduces confounding by ensuring treatment balance across the spatial domain[35]. However, the formal comparison of SRS and BSS within the context of spatial cluster randomized trials that incorporate both a spatial autoregressive (SAR) model for background dependence *and* an explicit spillover mechanism represents a novel contribution of the present work. To our knowledge, no prior study has examined how the choice between these two allocation strategies affects the bias–variance decomposition of the intervention effect estimator under SAR modeling with simultaneous spillover effects, nor has the

exhaustive enumeration of all feasible allocations been used to characterize the full design space of a spatial CRT.

A related line of work concerns covariate-constrained randomization, in which the set of eligible allocations is restricted to those achieving covariate balance across treatment arms[36]. While BSS and covariate-constrained randomization both restrict the allocation space, they operate on fundamentally different principles: BSS imposes a spatial isolation condition based on cluster adjacency (no two same-treatment clusters share a Rook boundary), whereas covariate-constrained randomization restricts allocations based on balance of measured covariates. These approaches are complementary and could in principle be combined; however, the present work focuses on the spatial isolation constraint as the design criterion, leaving covariate-balance extensions to future research.

Our findings show a clear relationship between the magnitude of spillover effects and the estimation error of the intervention effect. As the spillover effect (ψ) increases, the Mean Squared Error (MSE) for the intervention effect (β) generally rises, regardless of the sampling method or the presence of spatial dependence. This highlights the importance of accounting for spillover effects, as their increasing influence can diminish the accuracy of intervention effect estimates. ~~The results indicate that~~ On the 2×4 and 3×4 grids, randomly selected treatment assignments ~~lead to reduced average error (lower bias)~~ led to lower bias when estimating intervention effects, while block stratified treatment assignments ~~consistently result in reduced variation (lower variance) among a series of estimates~~resulted in less variation in MSE across allocations in most scenarios. On the 3×3 grid, whose six BSS allocations are not all checkerboards, BSS had the smaller bias in most scenarios, and under TrtSpill its MSE varied more across allocations than that of SRS. This suggests a trade-off: SRS provides a more accurate central estimate on average, whereas BSS offers greater precision and consistency across repeated trials, potentially with higher bias.

The study also clarifies the impact of spatial dependence (ρ) on intervention effect estimation error. The presence of even a small degree of spatial dependence (e.g., $\rho = 0.01$) ~~generally~~ increased MSE for ~~both sampling methods.~~ SRS in every scenario. For BSS, the effect depended on the grid: MSE rose on the 3×3 grid but fell for the checkerboard allocations of the 2×4 and 3×4 grids. This indicates that spatial correlation among observations can impede the accurate estimation of intervention effects if not managed during the design phase. The observed patterns across different spatial grid dimensions (2×4, 3×3, and 3×4) support these conclusions, demonstrating the consistency of the findings across varying

spatial configurations.

The investigation into cluster spillover restrictions (spillover to control clusters only vs. spillover to both control and intervention clusters) also identified important differences. When spillover was restricted to control clusters only and spatial dependence was present, the MSE for the intervention effect was higher for SRS ~~compared to BSS , particularly with increasing spillover magnitude~~than for BSS on every grid. Without spatial dependence, SRS had the lower MSE on the 2×4 and 3×4 grids and BSS on the 3×3 grid. Conversely, when spillover was permitted for both control and intervention clusters, SRS ~~often~~ exhibited lower MSE in every scenario, especially when spatial dependence was absent. This understanding of spillover application types provides investigators with a more precise basis for selecting a sampling strategy that aligns with their specific hypotheses regarding how an intervention's effects might propagate.

It is also insightful to note that for each simulation grid setup (2×4, 3×3, and 3×4), there are 7, 4, and 6 different treatment assignment combinations, respectively, each of which is the overall "optimal" combination for at least one of the simulation scenarios in terms of MSE of the intervention effect estimate. It appears that the optimal treatment combination shifts as the components of each simulation scenario are adjusted and there is no consistent treatment combination from the Simple Random Sampling strategy that is optimal. ~~This serves as further evidence to suggest that it is prudent to recommend the Block Stratified Sampling strategy because it provides results that consistently exhibit minimal MSE of the intervention effect estimate although the~~ The BSS treatment combinations are never the absolute best case in terms of MSE for any given simulation scenario~~. By implementing a BSS strategy, investigators will consistently obtain strong estimates of the intervention effect (although not the best for any specific single case) under the conditions of the spatial autoregressive (SAR) model~~, and BSS does not give the lowest average MSE in general. Its benefit is that it excludes the poorly balanced allocations that SRS can draw, which limits the worst case, most clearly when spillover flows into control clusters and spatial dependence is present.

~~An important nuance observed in the~~ Strict checkerboard allocations, which are the BSS allocations on the 2×4 and 3×4 ~~simulation results warrants attention~~ grids, warrant a further note in the context of practical design recommendations. ~~When BSS forces an exact checkerboardarrangement, maximally separating intervention~~ In a checkerboard, every control cluster borders an intervention cluster and

no intervention cluster borders another, so the spillover indicator equals $1 - x_i$ under both spillover types. The two spillover types then produce identical data, β and ~~control clusters, every intervention cluster is surrounded exclusively by control neighbors. Under TrtSpill, this means spillover flows from each intervention cluster into its control neighbors *and* back into neighboring intervention clusters (which are diagonally adjacent but not Rook neighbors~~ψ are not separately identifiable, and $\hat{\beta}$ absorbs the spillover effect, with bias near $-\psi$ when $\rho = 0$ (Section 5.2). This ~~structural property creates a systematic pattern of amplified spillover-induced bias specifically for BSS in the TrtSpill regime with spatial dependence, producing estimation error that exceeds that of SRS in that scenario. We refer to this as the "checkerboard paradox": the very design feature that makes BSS effective at minimizing spillover contamination under TrtNoSpill, complete spatial separation of treatment arms, becomes a liability when bidirectional spillover is expected~~is why the BSS results on these grids are identical under the two spillover types. Investigators should therefore consider their prior knowledge of the spillover mechanism when selecting between SRS and BSS, with BSS most advantageous when spillover is expected to flow primarily into control clusters.

In summary, this study offers a comparison of two opposing sampling methodologies in spatial CRTs. The patterns observed regarding spillover magnitude, spatial dependence, and spillover application types provide practical information for investigators. They must consider the goal of an unbiased average estimate (generally favored by SRS) versus the need for consistent, low-variance estimates (generally achieved with BSS), especially when designing studies where spillover effects are expected.

5.2 Limitations

While this simulation study provides valuable insights, it is important to acknowledge a few key limitations that may affect the generalizability and scope of its findings. As described in Section 3.2, in scenarios where spillover propagates only to adjacent control clusters, the spillover indicator z_i can become highly correlated with, or exactly collinear with, the intervention indicator x_i for certain allocation configurations. When this collinearity is present, $\hat{\beta}$ may absorb part of the spillover signal, producing systematic negative bias; this pattern reflects a structural property of the small grid geometries examined here, where the set of feasible allocations is limited, rather than a limitation of the estimation approach itself. The magnitude of this bias warrants candid discussion. With a true intervention effect of $\beta = 1.0$

and spillover magnitudes $\psi \in [0.5, 0.8]$, an exactly collinear allocation yields $\hat{\beta} \approx \beta - \psi$, so the bias of $\hat{\beta}$ is approximately $-\psi$, a distortion of 50–80% of the true effect, large enough to materially understate or even reverse a practical conclusion. We acknowledge that, in this regime, the estimation error can be severe. Two observations place that severity in context. First, because the ~~standard deviation of the block-stratified estimator is near zero~~variance of these collinear estimates is small relative to their squared bias, this error is almost entirely *bias* arising from lost identifiability rather than *variance*: it is a predictable consequence of the exact x_i–z_i confounding and afflicts any allocation that surrounds every control cluster with treated neighbors, not block stratification in particular, as the equally collinear Simple Random Sampling allocations exhibit comparable bias. Second, where the allocation instead permits the spillover term to be estimated separately, as in many Simple Random Sampling allocations under bidirectional spillover, the bias of $\hat{\beta}$ falls to roughly 1–2% of β (Table 1). The comparative, risk-management findings of this study are therefore unaffected by the magnitude of the absolute bias; rather, that magnitude motivates analysis-stage remedies that restore identifiability directly, which we discuss in Section 5.3. As discussed in Section 2.5, ρ (background spatial autocorrelation) and ψ (spillover) are conceptually distinct but both depend on the same neighborhood structure, creating an identification challenge when both are large; in such settings, estimates of $\hat{\psi}$ and $\hat{\rho}$ may exhibit instability, inflating MSE for both parameters. It is also necessary to note that the expanded simulation space quickly becomes computationally infeasible. The current study explored a specific range of spillover effect sizes, spatial dependence parameters, and grid dimensions. However, a more extensive exploration of additional parameters (e.g., different baseline effects, varying intervention effect sizes, or a wider range of spatial dependence values) or more complex spatial structures would require significantly greater computational resources. This limitation restricted the number of scenarios that could be thoroughly investigated within the current framework.

An additional notable limitation is the rudimentary spillover mechanism (binary effect) applied. This simulation study modeled spillover as a fixed, binary event, meaning a unit either experienced the full potential spillover effect or none, based solely on contiguity (Rook neighbors). This approach does not account for more realistic scenarios where spillover effects may decay with distance, or where cumulative exposure from multiple treated neighbors could lead to additive or non-linear effects. The absence of more sophisticated distance-based decay functions (e.g., linear, categorical, exponential, or Gaussian

decay) or mechanisms for additive spillover limits the direct applicability of these specific quantitative results to real-world situations where such nuanced spillover dynamics are likely. A further limitation is the cluster-level uniformity of spillover exposure. The current simulation assigns binary spillover at the cluster level: all subjects within a cluster either experience or do not experience spillover based solely on whether their cluster shares a Rook boundary with an intervention cluster. In practice, subjects near the boundary of a control cluster may experience stronger spillover than those near the center, while subjects near non-intervention boundaries may experience none at all. This within-cluster heterogeneity in spillover exposure is not captured by the current model, and future work could explore boundary-level or subject-level spillover mechanisms that account for the spatial position of subjects within clusters.

Because this study evaluates estimator properties through exhaustive enumeration of all possible treatment allocations, the findings characterize the *design-space* performance of each sampling strategy rather than the sampling distribution arising from a single randomization. A consequence of the block stratification condition is that BSS admits only a small number of valid allocations (2–6 across the grids studied), so it does not generate a randomization distribution and does not support permutation- or randomization-based inference. This does not constrain the Frequentist inference available to practitioners: model-based SAR maximum likelihood estimation conditions on the realized allocation and yields a valid sampling distribution for the intervention effect from a single fixed design, whether that design was obtained through SRS or BSS. The simulation results should accordingly be interpreted as describing how well each design tends to perform across the full space of feasible allocations for a given grid geometry, with the near-deterministic BSS allocation understood as a deliberate, risk-averse design choice rather than a limitation of inference.

The current specification models cluster-level outcomes and therefore does not estimate or account for intra-cluster correlation (ICC); incorporating individual-level outcomes with a cluster-specific random effect to capture within-cluster correlation is left to future work. The current model also did not consider higher-order neighbors beyond immediate (Rook) contiguity. The study also utilized homogeneous cluster sizes and regular grid shapes. Real-world clusters often exhibit irregular shapes and varying sizes, which could influence spillover dynamics and the efficiency of different sampling strategies. The current findings may not fully extend to such heterogeneous spatial arrangements. Furthermore, the study assumes a fixed treatment allocation ratio (primarily 1:1 or near 1:1). Varying the proportion of intervention to

control clusters could impact the statistical power and bias of intervention effect estimates, especially in the presence of spillovers. Finally, it is crucial to note that the results obtained and recommendation for a Block Stratified Sampling (BSS) strategy hold under the conditions of the spatial autoregressive (SAR) model and it cannot be guaranteed that the apparent benefits of BSS persist under the conditions of an alternative spatial model.

We also emphasize the intended scope of this work. Its goal is design guidance, identifying how investigators should allocate treatment and control clusters so as to minimize estimation error for the structural intervention coefficient β under the SAR model, rather than the identification of formal causal estimands. As developed in Section 2.5, β and ψ are structural model coefficients, not marginal potential-outcome contrasts, and a contrast such as $E[Y(0,\mathbf{1})] - E[Y(0,\mathbf{0})]$ is a derived function of the full parameter set and the spatial multiplier rather than any single coefficient. The SAR specification could in principle support explicitly causal interpretations under additional identifying assumptions; pursuing such causal-centered estimands, rather than the design-focused error comparison studied here, is a natural direction for future work.

5.3 Future Considerations

Building on this study's findings, several areas for future research can further improve our understanding of sampling design in spatial cluster randomized trials. A key next step involves incorporating additional spillover structures beyond the binary mechanism. Future simulations should explore various distance-based decay functions, such as linear, categorical/step-wise, exponential, or Gaussian decay. This would provide a more realistic representation of how intervention effects attenuate over space and allow for the identification of optimal sampling strategies under different decay profiles. For example, an intervention with rapid exponential decay might benefit from a different sampling approach than one with a more gradual linear decay.

Furthermore, the current model's assumption of non-additive spillover effects could be relaxed. Future research should consider the implications of "additive" spillover, where being a neighbor to multiple intervention cells allows for a greater cumulative spillover effect. This would reflect scenarios where the intensity of exposure to the intervention's indirect effects scales with the number or proximity of treated neighbors. Understanding how sampling methods perform under such additive spillover mechanisms is

important for interventions where collective exposure drives indirect effects. Exploring other neighbor structures beyond Rook contiguity is also warranted. Investigating Queen neighbors (sharing an edge or vertex) or higher-order neighbors (neighbors of neighbors) would provide a more complete understanding of how different definitions of "proximity" influence spillover propagation and, consequently, the performance of various sampling designs.

Additionally, future work could investigate the impact of "prior" weighting of intervention based on cluster subject density. In real-world settings, clusters rarely have uniform population density. Incorporating variable subject densities within clusters and exploring how this influences spillover effects and optimal sampling strategies would add realism to the simulations. An extension of the simulation parameters to cover a broader range of values for spillover effects, spatial dependence, and grid dimensions would allow for a more comprehensive mapping of the parameter space and a more robust set of recommendations for practitioners. This would enable a deeper understanding of the conditions under which each sampling method (SRS vs. BSS) is most advantageous, leading to more efficient and reliable designs for spatial cluster randomized trials.

Future research could also investigate the robustness of findings to model misspecification. This includes evaluating how different sampling strategies perform when alternative spatial models (e.g., Spatial Error Model, Spatial Durbin Model) are used for analysis, or when the true underlying spatial process deviates from the assumed SAR model. A natural model extension is to incorporate individual-level outcomes with a cluster-specific random effect, which would allow intra-cluster correlation (ICC) to be parameterized and estimated alongside the SAR spatial dependence and spillover coefficients. Finally, exploring optimal allocation ratios of intervention to control clusters in the presence of spillover effects and spatial dependence could provide valuable insights for maximizing statistical power or minimizing estimation error. Incorporating cost-effectiveness analysis into the design framework would also be beneficial, as different sampling strategies may have varying logistical and financial implications in real-world implementation.

A particularly promising direction is the evaluation of buffer-zone ("fried-egg") designs, in which control clusters bordering an intervention cluster are excluded from the primary analysis to reduce spillover contamination[8]. Such designs address the identifiability problem at its root: by removing the contaminated boundary controls, they restore $z_i = 0$ for the analyzed control clusters, breaking the exact

x_i–z_i collinearity by construction and allowing β and ψ to be estimated separately rather than conflated. In this respect, buffer zones target the source of the large biases characterized in Section 5.2, whereas the allocation-stage strategies studied here manage that error only through the bias–variance trade-off. A practical obstacle is that buffer-zone exclusion is not feasible on the compact grids examined in this study: under the block-stratified checkerboard, every control cluster borders an intervention cluster, so excluding the boundary controls would remove the entire control arm. Evaluating buffer-zone designs therefore requires larger spatial layouts in which interior clusters survive boundary exclusion. Characterizing the comparative performance of SRS, BSS, and buffer-zone exclusion on such layouts, especially ~~in the bidirectional-spillover-with-dependence scenarios where BSS exhibited amplified bias~~for allocations in which treatment and spillover exposure are collinear, would provide more complete guidance for practitioners and is an important next step in this line of research.

Declarations

Ethics Approval and Consent to Participate

Not applicable. This study involves simulation data only and does not require ethics approval or consent.

Consent for Publication

Not applicable.

Availability of Data and Materials

All data supporting the findings of this study are generated via simulation for the included simulation study. The simulation data and R code are available upon request and with permission of the authors.

Competing Interests

The authors declare that they have no competing interests.

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Authors' Contributions

A.W. conceptualized and designed the simulation study, wrote the manuscript text, and prepared Figures 1–7 and 10. T.V. provided Figures 8 and 9 and contributed advisement for Section 4 of the manuscript. F.-C.L. provided methodological guidance and critical review throughout. All authors reviewed and approved the final manuscript.

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Appendices

A Spatial Weights Matrix Formulation

The spatial weights matrix $\mathbf{W}$ encodes the presence and strength of spatial connections between observations i and j. Two types of weights are considered in this work.

Indicator-Based (Contiguity) Weights

Indicator-based weights are defined as

$$\tilde{w}_{ij} = \begin{cases} 1, & \text{if clusters } i \text{ and } j \text{ are Rook neighbors (shared edge)} \\ 0, & \text{otherwise} \end{cases}$$

and are row-normalized via

$$W_{ij} = \frac{\tilde{w}_{ij}}{\sum_{j \in \mathcal{N}(i)} \tilde{w}_{ij}}$$

so that each row of $\mathbf{W}$ sums to 1, making the spatial lag $\rho\mathbf{W}\mathbf{y}$ a weighted average of neighboring observations' responses.

Distance-Based Weights

Distance-based weights assign strength proportional to the inverse distance between cluster centroids:

$$\tilde{w}_{ij} = \begin{cases} \dfrac{1}{d_{ij}}, & \text{if } 0 < d_{ij} < d_{\max} \\ 0, & \text{otherwise} \end{cases}$$

where d_{ij} is the distance between clusters i and j and $d_{\max}$ is the maximum distance threshold for neighborhood inclusion. These weights are also row-normalized via

$$W_{ij} = \frac{\tilde{w}_{ij}}{\sum_{j \neq i} \tilde{w}_{ij}}.$$

In this study, indicator-based (Rook contiguity) weights are used throughout all simulation experiments.

B SAR Model: Full Likelihood

The cluster-level spatial lag model is

$$y_i = \alpha + \beta x_i + \psi z_i + \rho \sum_{j \in \mathcal{N}(i)} w_{ij} y_j + \varepsilon_i$$

with residuals

$$\varepsilon_i = y_i - (\alpha + \beta x_i + \psi z_i) - \rho \sum_{j \in \mathcal{N}(i)} w_{ij} y_j.$$

Assuming $\varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$, the likelihood function is

$$L\big(\alpha, \beta, \psi, \rho, \sigma^2\big) = \frac{|\mathbf{I} - \rho \mathbf{W}|}{(2\pi\sigma^2)^{n/2}} \exp\!\left(-\frac{1}{2\sigma^2}\, \boldsymbol{\varepsilon}^\top \boldsymbol{\varepsilon}\right)$$

where $|\mathbf{I} - \rho \mathbf{W}|$ is the Jacobian of the transformation from $\boldsymbol{\varepsilon}$ to $\mathbf{y}$. The log-likelihood is

$$\begin{aligned} \log L\big(\alpha, \beta, \psi, \rho, \sigma^2\big) = \log |\mathbf{I} - \rho \mathbf{W}| - \frac{n}{2} \log\big(2\pi\sigma^2\big) \\ - \frac{1}{2\sigma^2} \sum_{i=1}^{n} \left(y_i - \alpha - \beta x_i - \psi z_i - \rho \sum_{j \in \mathcal{N}(i)} w_{ij} y_j \right)^2 \end{aligned}$$

which is maximized with respect to $\alpha, \beta, \psi, \rho$, and σ^2. In practice, this maximization is performed by the `lagsarlm()` function from the **spdep** R library[28].

C Simulation Algorithm

The step-by-step simulation procedure used to generate results for all spatial grid geometries and scenarios is as follows.

1. **Define scenario.** Specify the spatial grid geometry and fix the simulation parameters ($\alpha = 0.2,\ \beta = 1.0,\ \psi \in \{0.5, 0.6, 0.7, 0.8\},\ \rho \in \{0.00, 0.01\}$) along with the spillover application type (TrtNoSpill or TrtSpill).

2. **Enumerate treatment combinations.** List all possible combinations of intervention and control cluster assignments for the specified grid: 70 for the 2×4 grid, 126 for the 3×3 grid, and 924 for the 3×4 grid.

3. **Construct spatial object.** Build the grid object (8, 9, or 12 clusters) using the **sf** R package[37] and define Rook neighbors for each cluster.

4. **Identify BSS allocations.** From the full set of combinations, identify those satisfying the block stratification condition: 2 for the 2×4 grid, 6 for the 3×3 grid, and 2 for the 3×4 grid.

5. **Simulate for each combination.** For each treatment combination:

   (a) Sample 20 subjects per cluster with uniformly distributed coordinates over the spatial extent (160, 180, or 240 subjects total).

   (b) Assign each subject to its cluster and intervention/control status.

   (c) Determine spillover eligibility: for each subject, set $z_i = 1$ if the subject's cluster has at least one Rook neighbor assigned to the intervention and the spillover application type permits it; otherwise $z_i = 0$.

   (d) Generate outcomes from the reduced (equilibrium) form (Equation 2): $\mathbf{y} = (\mathbf{I} - \rho\mathbf{W})^{-1}(\boldsymbol{\alpha} + \mathbf{X}\beta + \mathbf{Z}\psi + \boldsymbol{\varepsilon})$, where $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I})$ with $\sigma^2 = 0.01$.

   (e) Fit the spatial lag model via `lagsarlm()` and retain estimates $(\hat{\alpha}, \hat{\beta}, \hat{\psi}, \hat{\rho})$.

   (f) Repeat (a)–(e) $N = 50$ times for the 2×4 and 3×3 grids; $N = 10$ times for the 3×4 grid.

6. **Aggregate estimates.** For each treatment combination, compute the mean, variance, bias, and MSE of $\hat{\beta}$ (and other parameters) across the N replications.

7. **Summarize by design.** Aggregate metrics across all SRS combinations and separately across all BSS combinations to produce the summary statistics reported in Tables 1–3.

8. **Repeat.** Repeat Steps 1–5 for each grid geometry (2×4, 3×3, 3×4).

D Distance-Based Neighbor Construction

Distance-based neighbors[27] are constructed under the assumption that clusters within a specified distance of a reference cluster are considered neighbors. As illustrated in Figure 1 (c), the reference cell (dark) claims as neighbors all cells whose centroids lie within a specified radius $d_{\max}$.

Formally, cluster j is a distance-based neighbor of cluster i if

$$d_{ij} < d_{\max}$$

where d_{ij} is the Euclidean distance between the centroids of clusters i and j. Distance-based neighborhoods can also be defined for individual point observations rather than areal units; in that case, any point within $d_{\max}$ of the reference point is a neighbor. This definition generalizes Rook contiguity in the sense that, for a regular grid with appropriately chosen $d_{\max}$, the distance-based neighbors coincide with the Rook neighbors (or Queen neighbors if $d_{\max}$ is slightly larger). For irregular geographies, distance-based neighbors offer a flexible alternative that does not require shared boundaries. The corresponding distance-based spatial weights and their row normalization are given in Appendix A.